Permutations and Combinations: Questions and Full Solutions

Question 01

English

Eight students are to be split into study circles such that every circle has at least two students. The circles are not labelled, so two splittings are the same if they consist of the same circles. The number of splittings in which at least two of the circles have the same number of students is

(A) 715
(B) 665
(C) 630
(D) 1120

Only one option is correct.

हिन्दी

आठ विद्यार्थियों को अध्ययन-समूहों में इस प्रकार विभाजित करना है कि प्रत्येक समूह में कम से कम दो विद्यार्थी हों। समूहों पर कोई नाम या क्रमांक नहीं है, अतः दो विभाजन समान माने जाते हैं यदि उनके समूह समान हों। उन विभाजनों की संख्या, जिनमें कम से कम दो समूहों में विद्यार्थियों की संख्या समान है, है

(A) 715
(B) 665
(C) 630
(D) 1120

केवल एक विकल्प सही है।

Solution

A splitting is a partition of the 8 students into unlabelled groups, each of size at least 2. First list the possible multisets of group sizes, i.e. the partitions of 8 into parts ≥ 2:

one group: 8; two groups: $6+2,\ 5+3,\ 4+4$; three groups: $4+2+2,\ 3+3+2$; four groups: $2+2+2+2$ (five groups would need at least 10 students).

The condition 'at least two circles of the same size' holds exactly for the size patterns $4+4$, $4+2+2$, $3+3+2$ and $2+2+2+2$.

For unlabelled groups, the number of ways to divide n distinct objects into groups of sizes $a_1, a_2, \ldots$ is $\dfrac{n!}{a_1!\,a_2!\cdots}$ divided by $k!$ for every set of k groups of equal size, because permuting equal-sized groups gives the same splitting.

$4+4$: $\dfrac{8!}{4!\,4!}\cdot\dfrac{1}{2!}=\dfrac{70}{2}=35.$

$4+2+2$: $\dfrac{8!}{4!\,2!\,2!}\cdot\dfrac{1}{2!}=\dfrac{420}{2}=210.$

$3+3+2$: $\dfrac{8!}{3!\,3!\,2!}\cdot\dfrac{1}{2!}=\dfrac{560}{2}=280.$

$2+2+2+2$: $\dfrac{8!}{(2!)^4}\cdot\dfrac{1}{4!}=\dfrac{2520}{24}=105.$

Hence the required number is $35+210+280+105=630$. (For comparison, the patterns with all sizes different give $1+28+56=85$, and $630+85=715$ is the number of all splittings.)

Correct Option: (C)

Question 02

English

The eight letters of the word EQUATION are rearranged, each letter being used exactly once. The number of rearrangements in which none of the five vowels E, U, A, I, O occupies the position that it occupies in the word EQUATION is

(A) 21234
(B) 14833
(C) 264
(D) 15120

Only one option is correct.

हिन्दी

शब्द EQUATION के आठ अक्षरों को पुनर्व्यवस्थित किया जाता है, प्रत्येक अक्षर का ठीक एक बार प्रयोग करते हुए। उन पुनर्व्यवस्थाओं की संख्या, जिनमें पाँच स्वरों E, U, A, I, O में से कोई भी स्वर उस स्थान पर नहीं है जिस पर वह शब्द EQUATION में है, है

(A) 21234
(B) 14833
(C) 264
(D) 15120

केवल एक विकल्प सही है।

Solution

The vowels E, U, A, I, O occupy positions 1, 3, 4, 6, 7 of EQUATION. All $8! = 40320$ rearrangements are considered.

For a vowel x, let V_x be the set of rearrangements in which x is in its original position. We need the number of rearrangements in none of V_E, V_U, V_A, V_I, V_O.

If a particular set of k vowels is kept in place, the remaining $8-k$ letters can be arranged in $(8-k)!$ ways, so the intersection of any k of the sets has $(8-k)!$ elements.

By the principle of inclusion-exclusion, the required number is $\sum_{k=0}^{5}(-1)^k\binom{5}{k}(8-k)!$.

$= 8! - 5 \cdot 7! + 10 \cdot 6! - 10 \cdot 5! + 5 \cdot 4! - 3!$

$= 40320 - 25200 + 7200 - 1200 + 120 - 6 = 21234.$

This is a partial derangement: only the five vowel positions are forbidden, while the consonants Q, T, N may stay in their places. The full derangement number $D_8 = 14833$ would wrongly forbid the consonants too, and $D_5 \cdot 3! = 264$ would wrongly keep the vowels among the vowel positions.

Correct Option: (A)

Question 03

English

Six students $S_1, S_2, \ldots, S_6$ write a test. For peer review the six answer scripts are redistributed among them, one script to each student, in such a way that no student receives the script written by that student; all such redistributions are equally likely. The probability that S_1 receives the script of S_2 and S_2 receives the script of S_1 is

(A) $\dfrac{1}{30}$

(B) $\dfrac{9}{265}$

(C) $\dfrac{1}{25}$

(D) $\dfrac{44}{265}$

Only one option is correct.

हिन्दी

छः विद्यार्थी $S_1, S_2, \ldots, S_6$ एक परीक्षा देते हैं। परस्पर मूल्यांकन के लिए छः उत्तर-पुस्तिकाएँ उनमें इस प्रकार पुनः वितरित की जाती हैं कि प्रत्येक विद्यार्थी को एक पुस्तिका मिले और किसी भी विद्यार्थी को उसकी स्वयं लिखी पुस्तिका न मिले; ऐसे सभी वितरण समसम्भाव्य हैं। S_1 को S_2 की पुस्तिका तथा S_2 को S_1 की पुस्तिका मिलने की प्रायिकता है

(A) $\dfrac{1}{30}$

(B) $\dfrac{9}{265}$

(C) $\dfrac{1}{25}$

(D) $\dfrac{44}{265}$

केवल एक विकल्प सही है।

Solution

A redistribution is a permutation σ of $\{1, \ldots, 6\}$ (S_i receives the script of $S_{\sigma(i)}$) with $\sigma(i) \neq i$ for all i, i.e. a derangement. All derangements are equally likely, so the sample space has D_6 elements.

$D_n = n! \sum_{k=0}^{n} \dfrac{(-1)^k}{k!}$ gives $D_4 = 24\left(\dfrac{1}{2} - \dfrac{1}{6} + \dfrac{1}{24}\right) = 9$ and $D_6 = 720\left(\dfrac{1}{2} - \dfrac{1}{6} + \dfrac{1}{24} - \dfrac{1}{120} + \dfrac{1}{720}\right) = 360 - 120 + 30 - 6 + 1 = 265$.

Favourable outcomes: $\sigma(1) = 2$ and $\sigma(2) = 1$. Then σ maps $\{3, 4, 5, 6\}$ onto itself, and the condition $\sigma(i) \neq i$ for $i = 3, 4, 5, 6$ says this restriction is a derangement of four objects: $D_4 = 9$ ways.

Hence the probability is $\dfrac{9}{265}$.

(Treating the redistribution as an arbitrary permutation would give $\dfrac{4!}{6!} = \dfrac{1}{30}$, which ignores the condition that defines the sample space.)

Correct Option: (B)

Question 04

English

A row has 16 parking slots numbered $1, 2, \ldots, 16$. Four identical cars are to be parked, at most one car in a slot, such that slots 1 and 16 remain empty and between any two parked cars there are at least two empty slots. The number of ways of choosing the four slots is

(A) 126
(B) 210
(C) 495
(D) 70

Only one option is correct.

हिन्दी

एक पंक्ति में $1, 2, \ldots, 16$ क्रमांकित 16 पार्किंग स्थान हैं। चार सर्वसम कारें इस प्रकार खड़ी करनी हैं कि किसी स्थान पर अधिकतम एक कार हो, स्थान 1 तथा 16 खाली रहें, और किन्हीं दो खड़ी कारों के बीच कम से कम दो खाली स्थान हों। चार स्थानों को चुनने के तरीकों की संख्या है

(A) 126
(B) 210
(C) 495
(D) 70

केवल एक विकल्प सही है।

Solution

Choosing the slots is the same as choosing the numbers of empty slots before the first car (g_0), between consecutive cars (g_1, g_2, g_3) and after the last car (g_4). This correspondence is one-one because the cars are identical.

There are $16 - 4 = 12$ empty slots, so $g_0 + g_1 + g_2 + g_3 + g_4 = 12$.

Slot 1 empty means $g_0 \geq 1$, slot 16 empty means $g_4 \geq 1$, and at least two empty slots between cars means $g_1, g_2, g_3 \geq 2$.

Substitute $h_0 = g_0 - 1$, $h_4 = g_4 - 1$, $h_i = g_i - 2$ $(i = 1, 2, 3)$: $h_0 + \cdots + h_4 = 12 - 1 - 1 - 2 - 2 - 2 = 4$ with all $h \geq 0$.

By the beggar's (stars and bars) method, the number of non-negative solutions of a sum of 5 variables equal to 4 is $\binom{4+5-1}{5-1} = \binom{8}{4} = 70$.

Correct Option: (D)

Question 05

English

A committee of four persons is chosen at random from five married couples (ten persons). The probability that the committee contains at least two women and also contains at least one married couple is

(A) $\frac{10}{21}$

(B) $\frac{31}{42}$

(C) $\frac{11}{21}$

(D) $\frac{8}{21}$

Only one option is correct.

हिन्दी

पाँच विवाहित दम्पतियों (दस व्यक्तियों) में से चार व्यक्तियों की एक समिति यादृच्छया चुनी जाती है। समिति में कम से कम दो महिलाएँ होने तथा साथ ही कम से कम एक विवाहित दम्पति होने की प्रायिकता है

(A) $\frac{10}{21}$

(B) $\frac{31}{42}$

(C) $\frac{11}{21}$

(D) $\frac{8}{21}$

केवल एक विकल्प सही है।

Solution

Number of committees: $\binom{10}{4} = 210$, all equally likely.

Committees with at least two women: with w women and $4-w$ men, $\binom{5}{w}\binom{5}{4-w}$ ways. $w=2$: $10 \cdot 10 = 100$; $w=3$: $10 \cdot 5 = 50$; $w=4$: $5 \cdot 1 = 5$. Total 155.

Complementary counting inside this set: count those with no married couple. If the w women are fixed, a man may be chosen only if his wife is not among them, so the men come from the $5-w$ remaining husbands.

$w=2$: $\binom{5}{2}\binom{3}{2} = 10 \cdot 3 = 30$; $w=3$: $\binom{5}{3}\binom{2}{1} = 10 \cdot 2 = 20$; $w=4$: $\binom{5}{4}\binom{1}{0} = 5$. Total 55.

Hence committees with at least two women and at least one couple: $155 - 55 = 100$.

Probability $= \frac{100}{210} = \frac{10}{21}$.

Correct Option: (A)

Question 06

English

Let $n \geq 4$ be an integer such that

$$\sum_{k=0}^{3} \binom{3}{k}\binom{n}{k+4} = 5\binom{n+2}{6}.$$

Then n is equal to

(A) 28
(B) 32
(C) 35
(D) 38

Only one option is correct.

हिन्दी

माना कि $n \geq 4$ एक ऐसा पूर्णांक है कि

$$\sum_{k=0}^{3} \binom{3}{k}\binom{n}{k+4} = 5\binom{n+2}{6}.$$

तब n बराबर है

(A) 28
(B) 32
(C) 35
(D) 38

केवल एक विकल्प सही है।

Solution

Write $\binom{3}{k} = \binom{3}{3-k}$. Then the left side is $\sum_{k=0}^{3}\binom{3}{3-k}\binom{n}{k+4}$, a sum of products whose lower indices add up to $(3-k)+(k+4)=7$.

Vandermonde's identity $\sum_j \binom{a}{j}\binom{b}{r-j} = \binom{a+b}{r}$ (choose r objects from $a+b$ by splitting between the two groups) with $a=3$, $b=n$, $r=7$ gives LHS $= \binom{n+3}{7}$; the terms with $k+4>n$ or other out-of-range indices are zero on both sides.

Equivalently, apply Pascal's rule three times: $\binom{n}{r}+\binom{n}{r+1} = \binom{n+1}{r+1}$ turns the weights $1,3,3,1$ into $\binom{n+3}{7}$.

Using $\binom{m}{r} = \dfrac{m}{r}\binom{m-1}{r-1}$: $\binom{n+3}{7} = \dfrac{n+3}{7}\binom{n+2}{6}$.

For $n \geq 4$, $\binom{n+2}{6} \neq 0$, so the equation becomes $\dfrac{n+3}{7} = 5$, i.e. $n = 32$.

Check: $\binom{35}{7} = 6724520 = 5 \cdot 1344904 = 5\binom{34}{6}$.

Correct Option: (B)

Question 07

English

Let S be the set of 13 points consisting of the six vertices of a regular hexagon, the midpoints of its six sides and its centre. The number of triangles whose three vertices belong to S is

(A) 280
(B) 277
(C) 283
(D) 274

Only one option is correct.

हिन्दी

माना कि S उन 13 बिन्दुओं का समुच्चय है जिनमें एक सम षट्भुज के छः शीर्ष, उसकी छः भुजाओं के मध्य-बिन्दु तथा उसका

केन्द्र सम्मिलित हैं। उन त्रिभुजों की संख्या, जिनके तीनों शीर्ष S में हैं, है

(A) 280
(B) 277
(C) 283
(D) 274

केवल एक विकल्प सही है।

Solution

Three points of S form a triangle unless they are collinear, so the number of triangles is $\binom{13}{3}$ minus the number of collinear triples; $\binom{13}{3} = 286$.

Lines through three points of S:

(i) The six sides: each contains two vertices and the midpoint between them, giving 6 triples.

(ii) The three long diagonals: each joins opposite vertices and passes through the centre, giving 3 triples.

(iii) In a regular hexagon opposite sides are parallel and symmetric about the centre O, so the midpoints M and M' of opposite sides satisfy $\overrightarrow{OM'} = -\overrightarrow{OM}$; hence M, O, M' are collinear, giving 3 more triples.

No line contains four points of S, and no other triple is collinear: the vertices lie on one circle and the midpoints on a smaller concentric circle, so a line meets each circle in at most two points, and the lines through O that contain a vertex or a midpoint are exactly the six lines in (ii) and (iii). A line through a vertex and a midpoint of a non-adjacent side meets no third point of S (direct check with coordinates).

Collinear triples: $6 + 3 + 3 = 12$. Number of triangles: $286 - 12 = 274$.

Correct Option: (D)

Question 08

English

A path from $O(0,0)$ to the point $(6,6)$ consists of 12 unit steps, each step being one unit in the positive x-direction or one unit in the positive y-direction. The number of such paths that pass through none of the points $(1,2)$, $(2,1)$ and $(4,4)$ is

(A) 288
(B) 148
(C) 108
(D) 168

Only one option is correct.

हिन्दी

बिन्दु $O(0,0)$ से बिन्दु $(6,6)$ तक के एक पथ में 12 इकाई कदम होते हैं, प्रत्येक कदम धनात्मक x-दिशा में एक इकाई अथवा धनात्मक y-दिशा में एक इकाई का होता है। ऐसे पथों की संख्या, जो बिन्दुओं $(1,2)$, $(2,1)$ तथा $(4,4)$ में से किसी से भी नहीं गुजरते, है

(A) 288
(B) 148
(C) 108
(D) 168

केवल एक विकल्प सही है।

Solution

A monotone path from (a,b) to (c,d) with $c \geq a$, $d \geq b$ is an arrangement of $c-a$ x-steps and $d-b$ y-steps, so there are $\binom{(c-a)+(d-b)}{c-a}$ of them. All paths: $\binom{12}{6} = 924$.

Let A, B, C be the sets of paths through $P(1,2)$, $Q(2,1)$, $R(4,4)$.

$|A| = \binom{3}{1} \cdot \binom{5+4}{5} = 3 \cdot 126 = 378$; similarly $|B| = 378$; $|C| = \binom{8}{4} \cdot \binom{4}{2} = 70 \cdot 6 = 420$.

A path visits its points in increasing order of both coordinates. $P(1,2)$ and $Q(2,1)$ are not comparable ($1 < 2$ but $2 > 1$), so no path passes through both: $|A \cap B| = 0$, hence also $|A \cap B \cap C| = 0$.

$|A \cap C| = \binom{3}{1} \cdot \binom{3+2}{3} \cdot \binom{4}{2} = 3 \cdot 10 \cdot 6 = 180$, and likewise $|B \cap C| = 180$.

By inclusion-exclusion, $|A \cup B \cup C| = 378 + 378 + 420 - 0 - 180 - 180 + 0 = 816$.

Paths avoiding all three points: $924 - 816 = 108$. (A direct recursive count of paths, with the three points blocked, also gives 108.)

Correct Option: (C)

Question 09

English

The number of permutations σ of $\{1, 2, \ldots, 8\}$ having exactly two fixed points (that is, exactly two values of i with $\sigma(i) = i$), the two fixed points being consecutive integers, is

(A) 1855
(B) 7420
(C) 1484
(D) 2120

Only one option is correct.

हिन्दी

समुच्चय $\{1, 2, \ldots, 8\}$ के उन क्रमचयों σ की संख्या, जिनके ठीक दो स्थिर बिन्दु हैं (अर्थात् i के ठीक दो मानों के लिए $\sigma(i) = i$) तथा ये दोनों स्थिर बिन्दु क्रमागत पूर्णांक हैं, है

(A) 1855
(B) 7420
(C) 1484
(D) 2120

केवल एक विकल्प सही है।

Solution

A permutation with exactly the fixed points k and $k+1$ fixes these two numbers and moves every other number.

The pair of consecutive fixed points can be $\{1, 2\}, \{2, 3\}, \ldots, \{7, 8\}$: 7 choices.

For a chosen pair, σ permutes the remaining 6 numbers among themselves with no fixed point, i.e. as a derangement of 6 objects.

$D_6 = 6! \sum_{j=0}^{6} \frac{(-1)^j}{j!} = 720 - 720 + 360 - 120 + 30 - 6 + 1 = 265.$

Different pairs give different sets of fixed points, so there is no overlap. The number is $7 \cdot 265 = 1855$. (Without the consecutive condition it would be $\binom{8}{2} \cdot 265 = 7420$.)

Correct Option: (A)

Question 10

English

Let λ be a real number such that

$$\sum_{k=1}^{n} \left(k^2 + 3k + 1\right) k! = (n+2)! + (n+1)! + \lambda$$

for every positive integer n. Then the value of λ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि λ एक ऐसी वास्तविक संख्या है कि प्रत्येक धनात्मक पूर्णांक n के लिए

$$\sum_{k=1}^{n} \left(k^2 + 3k + 1\right) k! = (n+2)! + (n+1)! + \lambda$$

है। तब λ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Write the polynomial factor in falling-factorial form: $(k+1)(k+2) = k^2 + 3k + 2$, so $k^2 + 3k + 1 = (k+1)(k+2) - 1$.

Hence $(k^2 + 3k + 1)\, k! = (k+1)(k+2)\, k! - k! = (k+2)! - k!$.

Summing from $k = 1$ to n: $\sum_{k=1}^{n}(k+2)! - \sum_{k=1}^{n} k! = (3! + 4! + \cdots + (n+2)!) - (1! + 2! + \cdots + n!)$.

The factorials $3!, \ldots, n!$ occur in both sums and cancel, leaving $(n+1)! + (n+2)! - 1! - 2! = (n+2)! + (n+1)! - 3$.

This holds for every $n \geq 1$, so $\lambda = -3$.

Check with $n = 1$: left side $= (1 + 3 + 1) \cdot 1! = 5$; right side $= 3! + 2! - 3 = 5$. With $n = 2$: $5 + 11 \cdot 2 = 27 = 24 + 6 - 3$.

Final Answer: -3

Question 11

English

Let $N = \prod_{k=1}^{10} \frac{(2k)!}{k!}$. Then the greatest non-negative integer m such that 2^m divides N is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $N = \prod_{k=1}^{10} \frac{(2k)!}{k!}$ है। तब वह महत्तम अऋणात्मक पूर्णांक m, जिसके लिए 2^m, N को विभाजित करता है, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Split $(2k)!$ into its even and odd factors: $(2k)! = [2 \cdot 4 \cdot 6 \cdots (2k)] \cdot [1 \cdot 3 \cdot 5 \cdots (2k-1)]$.

The even part is $2^k(1 \cdot 2 \cdots k) = 2^k\, k!$. Hence $\dfrac{(2k)!}{k!} = 2^k \cdot [1 \cdot 3 \cdots (2k-1)]$, and the bracket is a product of odd numbers, so it is odd.

Therefore the exponent of 2 in $\dfrac{(2k)!}{k!}$ is exactly k.

The same follows from Legendre's formula: $e_2((2k)!) = \sum_{j\geq 1} \left[\dfrac{2k}{2^j}\right] = k + \sum_{j\geq 1} \left[\dfrac{k}{2^j}\right] = k + e_2(k!)$.
For example, $k = 10$: $e_2(20!) = 10 + 5 + 2 + 1 = 18$, $e_2(10!) = 5 + 2 + 1 = 8$, difference 10.
Exponents add over a product, so the exponent of 2 in N is $\sum_{k=1}^{10} k = 55$. Hence $m = 55$.

Final Answer: 55

Question 12

English

For a positive integer n, let $E(n)$ denote the greatest non-negative integer e such that 5^e divides $n!$. Then the value of $E(2026) - \dfrac{2026}{4}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

किसी धनात्मक पूर्णांक n के लिए माना कि $E(n)$ वह महत्तम अऋणात्मक पूर्णांक e है जिसके लिए 5^e, $n!$ को विभाजित करता है। तब $E(2026) - \dfrac{2026}{4}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

By Legendre's formula, the exponent of the prime 5 in $n!$ is $E(n) = \sum_{j\geq 1} \left[\dfrac{n}{5^j}\right]$.

For $n = 2026$: $\left[\dfrac{2026}{5}\right] = 405$, $\left[\dfrac{2026}{25}\right] = 81$, $\left[\dfrac{2026}{125}\right] = 16$, $\left[\dfrac{2026}{625}\right] = 3$, and $5^5 = 3125 > 2026$. So $E(2026) = 405 + 81 + 16 + 3 = 505$.
Then $E(2026) - \dfrac{2026}{4} = 505 - 506.5 = -1.5$.

Why the answer is negative: the formula can be written $E(n) = \dfrac{n - s_5(n)}{4}$, where $s_5(n)$ is the sum of the base-5 digits of n. Since $2026 = 3 \cdot 625 + 1 \cdot 125 + 1 \cdot 25 + 0 \cdot 5 + 1$, i.e. $(31101)_5$, $s_5(2026) = 6$ and $E(2026) - \dfrac{2026}{4} = -\dfrac{6}{4} = -1.5$, agreeing with the direct computation.

Final Answer: -1.50

Question 13

English

A four-digit number is formed using digits from the set $\{0, 1, 2, 3, 4\}$, repetition of digits being allowed and the first digit being non-zero. If one such number is chosen at random, then the probability that it is divisible by 11 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

समुच्चय $\{0, 1, 2, 3, 4\}$ के अंकों से एक चार अंकीय संख्या बनाई जाती है, जिसमें अंकों की पुनरावृत्ति की अनुमति है तथा प्रथम अंक शून्येतर है। यदि ऐसी कोई एक संख्या यादृच्छया चुनी जाए, तो उसके 11 से विभाज्य होने की प्रायिकता _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Numbers $\overline{abcd}$ with $a \in \{1, 2, 3, 4\}$ and $b, c, d \in \{0, 1, 2, 3, 4\}$: $4 \cdot 5^3 = 500$, all equally likely.

$\overline{abcd} = 1000a + 100b + 10c + d \equiv -a + b - c + d \pmod{11}$, so $11 \mid \overline{abcd}$ iff $11 \mid (a + c) - (b + d)$.

Here $0 \le a + c \le 8$ and $0 \le b + d \le 8$, so the difference lies in $[-8, 8]$ and must be 0: $a + c = b + d$.

Let $N(s)$ be the number of ordered pairs of digits from $\{0, \ldots, 4\}$ with sum s: for $s = 0, 1, \ldots, 8$ these are $1, 2, 3, 4, 5, 4, 3, 2, 1$.

For (a, c) we need $a \neq 0$; the pair $(0, s)$ is excluded when $s \le 4$, so the counts are $M(s) = 0, 1, 2, 3, 4, 4, 3, 2, 1$.

Favourable numbers: $\sum_{s=0}^{8} M(s)N(s) = 0 \cdot 1 + 1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + 4 \cdot 5 + 4 \cdot 4 + 3 \cdot 3 + 2 \cdot 2 + 1 \cdot 1 = 70$.

Probability $= \dfrac{70}{500} = \dfrac{7}{50} = 0.14$.

Final Answer: 0.14

Question 14

English

Let S be the set of all three-digit numbers whose digits are distinct, belong to $\{1, 2, 3, 4, 5, 6, 7\}$ and are in strictly increasing order from left to right. Then the arithmetic mean of the elements of S is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि S उन सभी तीन अंकीय संख्याओं का समुच्चय है जिनके अंक भिन्न हैं, $\{1, 2, 3, 4, 5, 6, 7\}$ में से हैं तथा बाएँ से दाएँ निरन्तर वर्धमान क्रम में हैं। तब S के अवयवों का समान्तर माध्य _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

A number in S is determined by its set of digits, written in increasing order, so $|S| = \binom{7}{3} = 35$, and its digits are the smallest, middle and largest element of a 3-subset.

Hundreds digit a: the other two digits are chosen from $\{a+1, \ldots, 7\}$, so a occurs $\binom{7-a}{2}$ times. Sum $= 1 \cdot 15 + 2 \cdot 10 + 3 \cdot 6 + 4 \cdot 3 + 5 \cdot 1 = 70$.

Tens digit b: one smaller digit from $b-1$ choices and one larger from $7-b$, so b occurs $(b-1)(7-b)$ times. Sum $= 2 \cdot 5 + 3 \cdot 8 + 4 \cdot 9 + 5 \cdot 8 + 6 \cdot 5 = 10 + 24 + 36 + 40 + 30 = 140$.

Units digit c: the other two from $\{1, \ldots, c-1\}$, so c occurs $\binom{c-1}{2}$ times. Sum $= 3 \cdot 1 + 4 \cdot 3 + 5 \cdot 6 + 6 \cdot 10 + 7 \cdot 15 = 210$.

Sum of all numbers in S: $100 \cdot 70 + 10 \cdot 140 + 210 = 7000 + 1400 + 210 = 8610$.

Mean $= \dfrac{8610}{35} = 246$. (Equivalently, the average smallest, middle and largest digits are $2, 4, 6$, giving 246; the reflection $d \mapsto 8 - d$ shows the average of the smallest and largest digits is 4.)

Final Answer: 246

Question 15

English

Let a be the number of three-digit numbers with distinct digits chosen from $\{1, 2, 3, 4, 5, 6, 7\}$ that are divisible by 9, and let b be the number of such three-digit numbers that are divisible by 25. Then the value of $\frac{a}{b}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि a, समुच्चय $\{1, 2, 3, 4, 5, 6, 7\}$ में से चुने गए भिन्न अंकों वाली उन तीन अंकीय संख्याओं की संख्या है जो 9 से विभाज्य हैं, तथा b ऐसी उन तीन अंकीय संख्याओं की संख्या है जो 25 से विभाज्य हैं। तब $\frac{a}{b}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Three distinct digits from $\{1, \ldots, 7\}$ have sum between $1+2+3=6$ and $5+6+7=18$. A number is divisible by 9 iff its digit sum is, so the digit sum must be 9 or 18.

Sum 9: $\{1, 2, 6\}$, $\{1, 3, 5\}$, $\{2, 3, 4\}$. Sum 18: only $\{5, 6, 7\}$. Each set of three distinct digits gives $3! = 6$ numbers, so $a = 4 \cdot 6 = 24$.

A number is divisible by 25 iff its last two digits form 00, 25, 50 or 75. Zero is not available, so the last two digits are 25 or 75 (distinct digits from the set).

For each ending, the hundreds digit is any of the 5 remaining digits, so $b = 2 \cdot 5 = 10$.

Hence $\frac{a}{b} = \frac{24}{10} = 2.4$.

Final Answer: 2.40

Question 16

English

Let T be the set of all positive integers having at most four digits, each digit being 1, 2 or 3. Then the arithmetic mean of the elements of T is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि T अधिकतम चार अंकों वाले उन सभी धनात्मक पूर्णांकों का समुच्चय है जिनका प्रत्येक अंक 1, 2 अथवा 3 है। तब T के अवयवों का समान्तर माध्य _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For each $k = 1, 2, 3, 4$ there are 3^k members of T with exactly k digits, so $|T| = 3+9+27+81 = 120$. Among the k-digit members, fixing a place and a digit leaves 3^{k-1} choices for the other places, so each of 1, 2, 3 appears 3^{k-1} times in each place. The digits in one place add up to $(1+2+3)3^{k-1} = 6 \cdot 3^{k-1}$.

Place values $1, 10, \ldots, 10^{k-1}$ form a GP with sum $\dfrac{10^k - 1}{9}$, so the sum of the k-digit members is $S_k = 6 \cdot 3^{k-1} \cdot \dfrac{10^k - 1}{9} = \dfrac{2}{3}\left(3^{k-1}10^k - 3^{k-1}\right)$.

Summing over k: $\sum_{k=1}^{4} 3^{k-1}10^k = 10\left(1 + 30 + 30^2 + 30^3\right) = 10 \cdot \dfrac{30^4 - 1}{30 - 1} = 10 \cdot \dfrac{809999}{29} = 279310$, a GP with ratio 30; and $\sum_{k=1}^{4} 3^{k-1} = \dfrac{3^4 - 1}{2} = 40$.

Sum of all members: $\dfrac{2}{3}(279310 - 40) = \dfrac{2}{3} \cdot 279270 = 186180$. (Check: $S_1 = 6$, $S_2 = 198$, $S_3 = 5994$, $S_4 = 179982$, total 186180.)

Mean $= \dfrac{186180}{120} = 1551.5$.

Final Answer: 1551.50

Question 17

English

All the distinct words (with or without meaning) obtained by arranging the six letters of the word PEPPER are listed in dictionary order. One of these words is chosen at random. Let p be the probability that the chosen word comes before PEPPER in the list and q the probability that it comes after PEPPER. Then the value of $p - q$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

शब्द PEPPER के छः अक्षरों को व्यवस्थित करके प्राप्त सभी भिन्न शब्दों (अर्थपूर्ण अथवा अर्थहीन) को शब्दकोश क्रम में सूचीबद्ध किया जाता है। इनमें से एक शब्द यादृच्छया चुना जाता है। माना कि p चुने गए शब्द के सूची में PEPPER से पहले आने की प्रायिकता है तथा q उसके PEPPER के बाद आने की प्रायिकता है। तब $p - q$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

PEPPER has the letters E, E, P, P, P, R, so the number of distinct words is $N = \dfrac{6!}{2!\,3!} = 60$. In dictionary order $E < P < R$.

Words beginning with E: the other letters E, P, P, P, R give $\dfrac{5!}{3!} = 20$ words, all before PEPPER.

Words beginning with P: next letter E (as in PEPPER). Third letter E: PEE followed by P, P, R gives $\dfrac{3!}{2!} = 3$ words, all before PEPPER.

Third letter P (as in PEPPER): fourth letter E gives PEPE followed by P, R, i.e. 2 words, before PEPPER; fourth letter P gives PEPP followed by E, R: PEPPER, PEPPRE, and PEPPER is the first of these.

So 25 words precede PEPPER and its rank is $r = 26$; the number of words after it is $60 - 26 = 34$.

$p = \dfrac{25}{60}$, $q = \dfrac{34}{60}$, so $p - q = \dfrac{2r - N - 1}{N} = -\dfrac{9}{60} = -0.15$.

Final Answer: -0.15

Question 18

English

The number of arrangements of all the letters of the word SUCCESSES that read the same from left to right as from right to left is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

शब्द SUCCESSES के सभी अक्षरों की उन व्यवस्थाओं की संख्या, जो बाएँ से दाएँ तथा दाएँ से बाएँ समान पढ़ी जाती हैं, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

SUCCESSES has 9 letters: S four times, C twice, E twice, U once.

In an arrangement that reads the same both ways, the letters in places i and $10 - i$ are equal for $i = 1, 2, 3, 4$, and place 5 is the middle.

So every letter outside place 5 is paired with an equal letter. U occurs once, so it cannot be paired and must be in place 5.

The remaining letters S^4, C^2, E^2 split equally between the halves: the left half (places 1 to 4) contains S, S, C, E, and it determines the right half.

The number of arrangements of S, S, C, E is $\dfrac{4!}{2!} = 12$, and each gives exactly one such arrangement of SUCCESSES. Hence the answer is 12.

Final Answer: 12

Question 19

English

Nine distinct books, namely three volumes of an encyclopaedia, three novels and three dictionaries, are arranged in a row on a shelf. The three volumes of the encyclopaedia are to be next to one another (in any order), no two novels are to be adjacent, and no novel is to be adjacent to a volume of the encyclopaedia. Then the number of such arrangements is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

नौ भिन्न पुस्तकें, अर्थात् एक विश्वकोश के तीन खण्ड, तीन उपन्यास तथा तीन शब्दकोश, एक शेल्फ पर एक पंक्ति में रखी जाती हैं। विश्वकोश के तीनों खण्ड एक-दूसरे के साथ (किसी भी क्रम में) रखने हैं, कोई दो उपन्यास आसन्न नहीं होने चाहिए, तथा कोई उपन्यास विश्वकोश के किसी खण्ड के आसन्न नहीं होना चाहिए। तब ऐसी व्यवस्थाओं की संख्या _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Bundle method: since the three volumes are together, treat them as one block B, which can be ordered internally in $3! = 6$ ways.

Arrange B and the three dictionaries: 4 units in a row, $4! = 24$ orders.

Gap method: the novels are placed in the 5 gaps (two ends and three between units). No two novels are adjacent exactly when each gap receives at most one novel.

A novel is adjacent to a volume exactly when it is in one of the two gaps on either side of B. If B is in the middle, these are two inner gaps; if B is at an end, they are the end gap and the gap after

B. In every case exactly 2 gaps are forbidden and 3 remain.

The three distinct novels must occupy the three allowed gaps, one in each: $3! = 6$ ways. Each such novel is then flanked only by dictionaries or the shelf end.

Total $= 3! \cdot 4! \cdot 3! = 6 \cdot 24 \cdot 6 = 864$.

Final Answer: 864

Question 20

English

From 4 men and 4 women, a team of 5 containing at least two women is chosen, and the members of the team are then seated in a row of 5 chairs. Let N_1 be the number of (team, seating) pairs in which no two women sit in adjacent chairs, and let N_2 be the number of (team, seating) pairs in which all the women of the team occupy consecutive chairs. Then the value of $N_1 - N_2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

4 पुरुषों तथा 4 महिलाओं में से कम से कम दो महिलाओं वाली 5 सदस्यों की एक टीम चुनी जाती है, और फिर टीम के सदस्यों को 5 कुर्सियों की एक पंक्ति में बैठाया जाता है। माना कि N_1 उन (टीम, बैठक-व्यवस्था) युग्मों की संख्या है जिनमें कोई दो महिलाएँ आसन्न कुर्सियों पर नहीं बैठतीं, तथा N_2 उन (टीम, बैठक-व्यवस्था) युग्मों की संख्या है जिनमें टीम की सभी महिलाएँ क्रमागत कुर्सियों पर बैठती हैं। तब $N_1 - N_2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Committee with constraint: a team has $w \in \{2, 3, 4\}$ women and $5 - w$ men. Number of teams: $w = 2$: $\binom{4}{2}\binom{4}{3} = 24$; $w = 3$: $\binom{4}{3}\binom{4}{2} = 24$; $w = 4$: $\binom{4}{4}\binom{4}{1} = 4$.

N_1 (gap method): seat the $5 - w$ men first, then put the women into distinct gaps among them. Viewing the seating as a row of M's and W's (alike objects), this is the condition that no two W's are adjacent.

$w = 2$: 3! orders of men, 4 gaps, ${}^4P_2 = 12$ placements: 72. $w = 3$: 2! orders of men, 3 gaps, ${}^3P_3 = 6$: 12. $w = 4$: one man gives only 2 gaps for 4 women, so no seating.

$N_1 = 24 \cdot 72 + 24 \cdot 12 + 4 \cdot 0 = 1728 + 288 = 2016$.

N_2 (bundle method): tie the w women into one block ($w!$ inner orders) and arrange the block with the $5 - w$ men: $(6 - w)!$ orders.

$w = 2$: $4! \cdot 2! = 48$; $w = 3$: $3! \cdot 3! = 36$; $w = 4$: $2! \cdot 4! = 48$. $N_2 = 24 \cdot 48 + 24 \cdot 36 + 4 \cdot 48 = 1152 + 864 + 192 = 2208$.

Hence $N_1 - N_2 = 2016 - 2208 = -192$.

Final Answer: -192

Question 21

English

The five letters of the word MOTTO are arranged in a row at random, all distinct arrangements being equally likely. Then the probability that no two identical letters are adjacent is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

शब्द MOTTO के पाँच अक्षरों को एक पंक्ति में यादृच्छया व्यवस्थित किया जाता है, सभी भिन्न व्यवस्थाएँ समसम्भाव्य हैं। तब किन्हीं दो सर्वसम अक्षरों के आसन्न न होने की प्रायिकता _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

MOTTO has M once and O, T twice each, so there are $\dfrac{5!}{2!\,2!} = 30$ distinct arrangements, all equally likely.

Gap method: first arrange M, T, T (3 distinct orders: TMT, TTM, MTT), then insert the two O's into the 4 gaps, at most one O per gap so that the O's are not adjacent.

Order TMT: the T's are separated; any 2 of the 4 gaps work: $\binom{4}{2} = 6$ arrangements.

Order TTM: the gaps are $_\,T\,_\,T\,_\,M\,_$. The T's are adjacent unless an O goes into the gap between them, so one O takes that gap and the other O takes one of the remaining 3 gaps: 3 arrangements. Order MTT: similarly 3.

Favourable arrangements: $6 + 3 + 3 = 12$. (Inclusion-exclusion check: $30 - 12 - 12 + 6 = 12$, where 12 arrangements have OO together, 12 have TT together and 6 have both.)

Probability $= \dfrac{12}{30} = \dfrac{2}{5} = 0.4$.

Final Answer: 0.40

Question 22

English

Seven distinct flowers, among them a rose R, a lily L and a tulip T, are used, all seven at a time. Let N_1 be the number of garlands that can be made in which R and L are adjacent and T is adjacent to neither R nor L. Let N_2 be the number of ways of placing the same seven flowers at seven equally spaced points around a fixed circular tray, with the same two conditions, where placements obtained from one another by a rotation are regarded as the same and clockwise and anticlockwise orders are regarded as different. Then the value of $N_1 - N_2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

सात भिन्न फूल, जिनमें एक गुलाब R, एक कुमुदिनी L तथा एक ट्यूलिप T हैं, सभी एक साथ प्रयुक्त किए जाते हैं। माना कि N_1 उन मालाओं की संख्या है जिनमें R तथा L आसन्न हैं तथा T न तो R के और न ही L के आसन्न है। माना कि N_2 इन्हीं सात फूलों को एक स्थिर वृत्ताकार थाल के चारों ओर सात समान दूरी वाले बिन्दुओं पर इन्हीं दोनों प्रतिबंधों के साथ रखने के तरीकों की संख्या है, जहाँ एक-दूसरे से घूर्णन द्वारा प्राप्त व्यवस्थाएँ समान मानी जाती हैं तथा दक्षिणावर्त और वामावर्त क्रम भिन्न माने जाते हैं। तब $N_1 - N_2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Circular tray, rotations identified: tie R and L into a block (RL or LR, 2 ways). With the other 5 flowers there are 6 units around the circle.

Fix the position of the block to remove rotations. The other 5 units occupy, in order going round, places $1, 2, 3, 4, 5$ after the block; places 1 and 5 are next to the block.

T is adjacent to neither R nor L exactly when T is in place 2, 3 or 4: 3 choices. The remaining 4 flowers fill the other places in $4! = 24$ ways.

$N_2 = 2 \cdot 3 \cdot 24 = 144$.

Garland: a garland can be turned over, so an arrangement and its reflection (clockwise and anti-clockwise orders) give the same garland. The conditions are preserved by reflection, and since the flowers are distinct no arrangement equals its own reflection. Hence each garland corresponds to exactly 2 tray arrangements and $N_1 = \dfrac{144}{2} = 72$.

$N_1 - N_2 = 72 - 144 = -72$.

Final Answer: -72

Question 23

English

Six distinct beads, of which three are red and three are blue, are strung together to form a necklace, all distinct necklaces being equally likely. Let p be the probability that red and blue beads alternate on the necklace, and let q be the probability that the three red beads are consecutive on the necklace. Then the value of $p - q$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

छः भिन्न मनकों को, जिनमें तीन लाल तथा तीन नीले हैं, एक साथ पिरोकर एक हार बनाया जाता है, सभी भिन्न हार समसम्भाव्य हैं। माना कि p हार पर लाल तथा नीले मनकों के एकान्तर क्रम में होने की प्रायिकता है, तथा q तीनों लाल मनकों के हार पर क्रमागत होने की प्रायिकता है। तब $p - q$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

A necklace can be rotated and turned over. For n distinct beads, there are $(n-1)!$ circular arrangements up to rotation, and turning over pairs them (no arrangement of distinct beads with $n \geq 3$ is its own mirror image), so there are $\dfrac{(n-1)!}{2}$ necklaces. For $n = 6$: $\dfrac{120}{2} = 60$.

Both properties (alternating colours, reds consecutive) are unchanged by rotation and by turning over, so in each case the number of necklaces is half the number of circular arrangements up to rotation.

Alternating: fix one red bead to remove rotations; the other two reds must be at the positions 2 and 4 places away (2! orders), and the blues fill the remaining 3 positions (3!). That gives 12 arrangements and 6 necklaces.

Reds consecutive: treat the reds as a block (3! inner orders); with the 3 blues there are 4 units around a circle, $(4-1)! = 6$ arrangements. That gives 36 arrangements and 18 necklaces.

$p = \dfrac{6}{60} = 0.1$ and $q = \dfrac{18}{60} = 0.3$, so $p - q = -0.2$.

Final Answer: -0.20

Question 24

English

Five married couples sit at a round table with ten equally spaced seats so that men and women alternate. Two seatings are regarded as the same if one can be obtained from the other by a rotation. Let Mr. X and Mrs. X be one of the couples. Then the number of seatings in which Mrs. X sits diametrically opposite Mr. X is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

पाँच विवाहित दम्पति दस समान दूरी वाली सीटों वाली एक गोल मेज़ पर इस प्रकार बैठते हैं कि पुरुष तथा महिलाएँ एकान्तर क्रम में हों। दो बैठक-व्यवस्थाएँ समान मानी जाती हैं यदि एक को दूसरे से घूर्णन द्वारा प्राप्त किया जा सके। माना कि श्री X तथा श्रीमती X इन दम्पतियों में से एक हैं। तब उन बैठक-व्यवस्थाओं की संख्या, जिनमें श्रीमती X, श्री X के ठीक व्यासतः सम्मुख बैठती हैं, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Number the seats $0, 1, \ldots, 9$ round the table. Fixing Mr. X at seat 0 removes the rotations, so each seating is counted once.

Men and women alternate, so the men occupy the even seats $0, 2, 4, 6, 8$ and the women the odd seats $1, 3, 5, 7, 9$.

The seat diametrically opposite seat 0 is seat 5. It is an odd seat, so Mrs. X can sit there and the alternation is kept: 1 way.

The other four men fill seats $2, 4, 6, 8$ in $4! = 24$ ways, and the other four women fill seats $1, 3, 7, 9$ in $4! = 24$ ways.

Number of seatings $= 24 \cdot 24 = 576$. (Without the condition on Mrs. X the alternating seatings number $4! \cdot 5! = 2880$.)

Final Answer: 576

Question 25

English

A bag contains three identical white balls and three balls of different colours (one red, one green and one blue). Let N_1 be the number of ways of selecting three balls from the bag. Let N_2 be the number of ways of selecting three balls if the three white balls are numbered 1, 2, 3, so that all six balls are distinguishable. Then the value of $\frac{N_1}{N_2}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक थैले में तीन सर्वसम सफ़ेद गेंदें तथा भिन्न रंगों की तीन गेंदें (एक लाल, एक हरी तथा एक नीली) हैं। माना कि N_1 थैले में से तीन गेंदें चुनने के तरीकों की संख्या है। माना कि N_2 तीन गेंदें चुनने के तरीकों की संख्या है यदि तीनों सफ़ेद गेंदों पर 1, 2, 3 अंकित हों, जिससे सभी छः गेंदें विभेद्य हो जाएँ। तब $\frac{N_1}{N_2}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

With identical white balls, a selection is fixed by which coloured balls it contains; if it contains k coloured balls ($\binom{3}{k}$ ways), it contains $3-k$ white balls, and there is only one way to take them because they are identical (and $3-k \le 3$ are available).

$N_1 = \binom{3}{0} + \binom{3}{1} + \binom{3}{2} + \binom{3}{3} = 1+3+3+1 = 8.$

With all six balls distinguishable, $N_2 = \binom{6}{3} = 20.$

$\frac{N_1}{N_2} = \frac{8}{20} = 0.4.$

Final Answer: 0.40

Question 26

English

A basket contains 4 identical apples, 2 identical bananas and 5 mangoes of five different varieties. Consider all possible selections of fruits from the basket, the empty selection included; two selections are different if they differ in the number of apples, the number of bananas or the set of mangoes chosen. Then the arithmetic mean of the number of fruits in a selection, taken over all these selections, is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक टोकरी में 4 सर्वसम सेब, 2 सर्वसम केले तथा पाँच भिन्न किस्मों के 5 आम हैं। टोकरी से फलों के सभी सम्भव चयनों पर विचार कीजिए, जिसमें रिक्त चयन भी सम्मिलित है; दो चयन भिन्न माने जाते हैं यदि उनमें सेबों की संख्या, केलों की संख्या अथवा चुने गए आमों का समुच्चय भिन्न हो। तब इन सभी चयनों पर लिए गए, एक चयन में फलों की संख्या का समान्तर माध्य _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

A selection is given by the number $a \in \{0, \ldots, 4\}$ of apples, the number $b \in \{0, 1, 2\}$ of bananas and a subset M of the 5 mangoes. Hence there are $5 \cdot 3 \cdot 2^5 = 480$ selections (including the empty one; 479 non-empty, the $(p+1)(q+1)2^n - 1$ count).

The size of a selection is $a + b + |M|$. Summing over all 480 selections, each value of a occurs $3 \cdot 32 = 96$ times, so $\sum a = 96(0+1+2+3+4) = 960$; each value of b occurs $5 \cdot 32 = 160$ times, so $\sum b = 160(0+1+2) = 480$.

For the mangoes, the number of subsets of size k is $\binom{5}{k}$ and each is combined with 15 choices of (a, b): $\sum |M| = 15 \sum_{k=0}^{5} k\binom{5}{k} = 15 \cdot 5 \cdot 2^4 = 1200$.

Total of the sizes $= 960 + 480 + 1200 = 2640$.

Mean $= \dfrac{2640}{480} = 5.5$.

Final Answer: 5.50

Question 27

English

Let $X = \{1, 2, 3\}$ and let $P(X)$ denote the power set of X. Then the number of ordered triples (A, B, C) of elements of $P(X)$ such that $A \subsetneq B \subsetneq C$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $X = \{1, 2, 3\}$ है तथा $P(X)$, X का घात समुच्चय है। तब $P(X)$ के अवयवों के उन क्रमित त्रिकों (A, B, C) की संख्या, जिनके लिए $A \subsetneq B \subsetneq C$ है, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

First count triples with $A \subseteq B \subseteq C$. Such a triple is fixed by saying, for each element of X, which of the four regions it lies in: A, $B \setminus A$, $C \setminus B$ or $X \setminus C$. So there are $4^3 = 64$ such triples.

Remove those with $A = B$ or $B = C$ by inclusion-exclusion. Triples with $A = B$ correspond to pairs $B \subseteq C$: each element in B, in $C \setminus B$ or outside, $3^3 = 27$. Likewise 27 triples have $B = C$.

Triples with $A = B = C$ correspond to single subsets: $2^3 = 8$.

Strict chains: $64 - 27 - 27 + 8 = 18$.

Check by sizes: $(|A|, |B|, |C|) = (0, 1, 2)$: $3 \cdot 2 = 6$; $(0, 1, 3)$: 3; $(0, 2, 3)$: 3; $(1, 2, 3)$: $3 \cdot 2 = 6$; total 18.

Final Answer: 18

Question 28

English

Consider all ordered quadruples (x_1, x_2, x_3, x_4) of integers satisfying

$$x_1 + x_2 + x_3 + x_4 = 12, \qquad x_1 \geq 1, \quad x_2 \geq 2, \quad 0 \leq x_3 \leq 5, \quad x_4 \geq 0.$$

Then the arithmetic mean of x_3 over all such quadruples is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

उन सभी क्रमित चतुष्कों (x_1, x_2, x_3, x_4) पर विचार कीजिए जिनके अवयव पूर्णांक हैं तथा

$$x_1 + x_2 + x_3 + x_4 = 12, \qquad x_1 \geq 1, \quad x_2 \geq 2, \quad 0 \leq x_3 \leq 5, \quad x_4 \geq 0$$

को सन्तुष्ट करते हैं। तब ऐसे सभी चतुष्कों पर x_3 का समान्तर माध्य _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Shift the lower bounds: $y_1 = x_1 - 1 \geq 0$, $y_2 = x_2 - 2 \geq 0$. The equation becomes $y_1 + y_2 + x_3 + x_4 = 9$ with all variables non-negative and $x_3 \leq 5$.

Fix $x_3 = t$ with $0 \leq t \leq 5$. Then $y_1 + y_2 + x_4 = 9 - t$ has $\binom{(9-t)+2}{2} = \binom{11-t}{2}$ non-negative solutions (beggar's method).

$t = 0, 1, 2, 3, 4, 5$ give $55, 45, 36, 28, 21, 15$ solutions. Total number of quadruples: 200. (Check by inclusion-exclusion: $\binom{12}{3} - \binom{6}{3} = 220 - 20 = 200$, removing $x_3 \geq 6$.)

Sum of x_3 over all quadruples: $0{\cdot}55{+}1{\cdot}45{+}2{\cdot}36{+}3{\cdot}28{+}4{\cdot}21{+}5{\cdot}15 = 0{+}45{+}72{+}84{+}84{+}75 = 360$.
Mean $= \dfrac{360}{200} = 1.8$.

Final Answer: 1.80

Question 29

English

Let a_k denote the coefficient of x^k in the expansion of $\left(1 + x - \dfrac{x^2}{2}\right)^5$. Then the value of a_4 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $\left(1 + x - \dfrac{x^2}{2}\right)^5$ के प्रसार में x^k का गुणांक a_k है। तब a_4 का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

By the multinomial theorem, $\left(1 + x - \dfrac{x^2}{2}\right)^5 = \sum \dfrac{5!}{p!\,q!\,r!}\, x^q \left(-\dfrac{1}{2}\right)^r x^{2r}$ over non-negative p, q, r with $p + q + r = 5$.
The power of x is $q + 2r$, so a_4 collects the triples with $q + 2r = 4$: counting these is the extraction step. $r = 0$: $q = 4, p = 1$; $r = 1$: $q = 2, p = 2$; $r = 2$: $q = 0, p = 3$.
$(1, 4, 0)$: $\dfrac{120}{1 \cdot 24 \cdot 1} = 5$.
$(2, 2, 1)$: $\dfrac{120}{2 \cdot 2 \cdot 1} \cdot \left(-\dfrac{1}{2}\right) = 30 \cdot \left(-\dfrac{1}{2}\right) = -15$.
$(3, 0, 2)$: $\dfrac{120}{6 \cdot 1 \cdot 2} \cdot \dfrac{1}{4} = 10 \cdot \dfrac{1}{4} = 2.5$.
$a_4 = 5 - 15 + 2.5 = -7.5$. (Check: writing the base as $1 + u$, $u = x\left(1 - \dfrac{x}{2}\right)$, gives $\binom{5}{2} \cdot \dfrac{1}{4} + \binom{5}{3}\left(-\dfrac{3}{2}\right) + \binom{5}{4} = 2.5 - 15 + 5$.)

Final Answer: -7.50

Question 30

English

The coefficient of x^7 in the expansion of $(1-x)^2\left(1+x+x^2\right)^6$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$(1-x)^2\left(1+x+x^2\right)^6$ के प्रसार में x^7 का गुणांक _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Since $(1-x)(1+x+x^2) = 1-x^3$, we have $(1-x)^2(1+x+x^2)^6 = (1-x^3)^2(1+x+x^2)^4 = (1-2x^3+x^6)(1+x+x^2)^4$.

Let c_k be the coefficient of x^k in $(1+x+x^2)^4$. By the multinomial theorem, $c_k = \sum \dfrac{4!}{p!\,q!\,r!}$ over non-negative p, q, r with $p+q+r=4$ and $q+2r=k$ (the term is $1^p x^q (x^2)^r$).

c_7: $q+2r=7$, $p+q+r=4$ gives only $(p,q,r)=(0,1,3)$: $\dfrac{4!}{0!\,1!\,3!} = 4$.

c_4: $(0,4,0)$: 1; $(1,2,1)$: $\dfrac{24}{1\cdot 2\cdot 1} = 12$; $(2,0,2)$: $\dfrac{24}{2\cdot 1\cdot 2} = 6$; so $c_4 = 19$.

c_1: $(3,1,0)$: $\dfrac{24}{6} = 4$.

The coefficient of x^7 is $1\cdot c_7 - 2\cdot c_4 + 1\cdot c_1 = 4 - 38 + 4 = -30$. (The full row of $(1+x+x^2)^4$ is $1, 4, 10, 16, 19, 16, 10, 4, 1$, which sums to $3^4 = 81$.)

Final Answer: -30

Question 31

English

Four distinct balls are placed in three distinct boxes, each box being allowed to remain empty. The number of placements in which at least two of the boxes contain the same number of balls (an empty box contains 0 balls) is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

चार भिन्न गेंदों को तीन भिन्न डिब्बों में रखा जाता है, प्रत्येक डिब्बे को रिक्त रहने की अनुमति है। उन व्यवस्थाओं की संख्या, जिनमें कम से कम दो डिब्बों में गेंदों की संख्या समान है (रिक्त डिब्बे में 0 गेंदें हैं), _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Each ball can go into any of the 3 boxes, so there are $3^4 = 81$ placements.

Complementary counting: count the placements in which the three boxes contain three different numbers of balls. Three different non-negative integers with sum 4 must be $0, 1, 3$ (since $0+1+2 = 3 < 4$ and any other choice with a 2 or a 4 exceeds 4 or repeats).

Assign the counts $0, 1, 3$ to the three distinct boxes: $3! = 6$ ways. Divide the 4 distinct balls into groups of unequal sizes 3, 1, 0: $\dfrac{4!}{3!\,1!\,0!} = 4$ ways. So $6 \cdot 4 = 24$ placements have all counts different.

Hence the required number is $81 - 24 = 57$.

Direct check by patterns with a repeated count: $(2, 1, 1)$: 3 choices of the box with 2, then $\binom{4}{2} \cdot 2! = 12$: 36; $(2, 2, 0)$: 3 choices of the empty box, then $\binom{4}{2} = 6$: 18; $(4, 0, 0)$: 3. Total $36 + 18 + 3 = 57$.

Final Answer: 57

Question 32

English

Let a be the number of ways of placing 5 distinct balls in 3 distinct boxes so that no box is empty, and let b be the number of ways of placing 5 identical balls in 3 distinct boxes so that no box is empty. Then the value of $\frac{b}{a}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि a, 5 भिन्न गेंदों को 3 भिन्न डिब्बों में इस प्रकार रखने के तरीकों की संख्या है कि कोई डिब्बा रिक्त न रहे, तथा b, 5 सर्वसम गेंदों को 3 भिन्न डिब्बों में इस प्रकार रखने के तरीकों की संख्या है कि कोई डिब्बा रिक्त न रहे। तब $\frac{b}{a}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Distinct balls: each ball has 3 choices, $3^5 = 243$ placements. By inclusion-exclusion, those with at least one empty box number $\binom{3}{1}2^5 - \binom{3}{2}1^5 = 96 - 3 = 93$, so $a = 243 - 93 = 150$.

(Check by size patterns: $(3,1,1)$: $3 \cdot \binom{5}{3} \cdot 2 = 60$; $(2,2,1)$: $3 \cdot 5 \cdot \binom{4}{2} = 90$; total 150.)

Identical balls: only the numbers $x_1, x_2, x_3 \geq 1$ in the boxes matter, with $x_1 + x_2 + x_3 = 5$. Putting $y_i = x_i - 1$: $y_1 + y_2 + y_3 = 2$, which has $\binom{4}{2} = 6$ solutions. So $b = 6$.

$$\frac{b}{a} = \frac{6}{150} = 0.04.$$

Final Answer: 0.04

Question 33

English

Fifteen identical sweets are distributed among four children A, B, C and D (a child may receive no sweet). Then the number of distributions in which A receives more sweets than B is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

पन्द्रह सर्वसम मिठाइयाँ चार बच्चों A, B, C तथा D में बाँटी जाती हैं (किसी बच्चे को कोई मिठाई न मिले, यह सम्भव है)। तब उन वितरणों की संख्या, जिनमें A को B से अधिक मिठाइयाँ मिलती हैं, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

A distribution is a solution of $a+b+c+d = 15$ in non-negative integers (the shares of A, B, C, D). By the beggar's method there are $\binom{15+3}{3} = \binom{18}{3} = 816$ distributions.

Swapping the shares of A and B is a one-one correspondence between distributions with $a > b$ and those with $a < b$, so these two sets have the same size.

Distributions with $a = b = k$: then $c + d = 15 - 2k$, which has $16 - 2k$ solutions, for $k = 0, 1, \ldots, 7$. Total ties: $16 + 14 + 12 + 10 + 8 + 6 + 4 + 2 = 72$.

Hence the number with $a > b$ is $\frac{816 - 72}{2} = \frac{744}{2} = 372$.

Final Answer: 372

Question 34

English

The number of functions $f : \{1, 2, 3, 4\} \to \{1, 2, 3\}$ whose range contains both 1 and 2 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

उन फलनों $f : \{1, 2, 3, 4\} \to \{1, 2, 3\}$ की संख्या, जिनके परिसर में 1 तथा 2 दोनों हैं, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

There are $3^4 = 81$ functions from a 4-element set to a 3-element set.

Let P be the set of functions whose range does not contain 1 and Q the set whose range does not contain 2. A function in P maps into $\{2, 3\}$: $|P| = 2^4 = 16$; likewise $|Q| = 16$; and $|P \cap Q| = 1$ (the constant function 3).

By inclusion-exclusion, the functions whose range contains both 1 and 2 number $81 - (16 + 16 - 1) = 50$.

Check: such a range is $\{1, 2\}$ or $\{1, 2, 3\}$. Onto functions to $\{1, 2, 3\}$: $3^4 - 3 \cdot 2^4 + 3 \cdot 1^4 = 36$; functions with range exactly $\{1, 2\}$: $2^4 - 2 = 14$; total 50.

Final Answer: 50

Question 35

English

Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{1, 2, 3\}$. For an onto function $f : A \to B$, the number of functions $g : B \to A$ such that $f(g(b)) = b$ for every $b \in B$ is denoted by $N(f)$. Then the arithmetic mean of $N(f)$ over all onto functions $f : A \to B$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $A = \{1, 2, 3, 4, 5\}$ तथा $B = \{1, 2, 3\}$ है। किसी आच्छादक फलन $f : A \to B$ के लिए, उन फलनों $g : B \to A$ की संख्या, जिनके लिए प्रत्येक $b \in B$ के लिए $f(g(b)) = b$ है, $N(f)$ द्वारा निरूपित की जाती है। तब सभी आच्छादक फलनों $f : A \to B$ पर $N(f)$ का समान्तर माध्य _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For a fixed onto f, the condition $f(g(b)) = b$ means $g(b)$ is an element of the fibre $f^{-1}(b)$, chosen independently for $b = 1, 2, 3$. Hence $N(f) = |f^{-1}(1)|\,|f^{-1}(2)|\,|f^{-1}(3)|$. The fibres are disjoint, so every such g is one-one (and it can never be onto, since $|B| < |A|$).

Number of onto f: by inclusion-exclusion $3^5 - \binom{3}{1}2^5 + \binom{3}{2}1^5 = 243 - 96 + 3 = 150$.

The fibre sizes are positive and add to 5, so they are $3, 1, 1$ or $2, 2, 1$ in some order.

Sizes $3, 1, 1$: choose the value with 3 preimages (3 ways), its preimages ($\binom{5}{3} = 10$), then the other two elements go to the other two values (2!): 60 functions, each with $N(f) = 3$.

Sizes $2, 2, 1$: choose the value with 1 preimage (3 ways) and that preimage (5 ways), then split the other 4 elements between the remaining two values ($\binom{4}{2} = 6$): 90 functions, each with $N(f) = 4$. (Check: $60 + 90 = 150$.)

$$\text{Mean} = \frac{60 \cdot 3 + 90 \cdot 4}{150} = \frac{180 + 360}{150} = \frac{540}{150} = 3.6.$$

Final Answer: 3.60

Question 36

English

A function is chosen at random from the set of all onto functions $f : \{1, 2, 3, 4, 5\} \to \{1, 2, 3, 4\}$. Then the probability that $f(i) \neq i$ for every $i \in \{1, 2, 3, 4\}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

सभी आच्छादक फलनों $f : \{1, 2, 3, 4, 5\} \to \{1, 2, 3, 4\}$ के समुच्चय में से एक फलन यादृच्छया चुना जाता है। तब प्रत्येक $i \in \{1, 2, 3, 4\}$ के लिए $f(i) \neq i$ होने की प्रायिकता _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Onto functions from a 5-set to a 4-set: $\sum_{m=0}^{4}(-1)^m\binom{4}{m}(4-m)^5 = 1024 - 972 + 192 - 4 = 240$.

Favourable functions are onto and satisfy the derangement-type condition $f(i) \neq i$ for $i = 1, 2, 3, 4$ ($f(5)$ is unrestricted).

Use inclusion-exclusion on the values not attained. For a set M of m values, count the functions with $f(i) \neq i$ ($i \leq 4$) whose values avoid M: for $i \in M$ the value $f(i)$ lies in the $4 - m$ allowed values (none equals i, as $i \in M$); for $i \leq 4$, $i \notin M$, there are $4 - m - 1 = 3 - m$ allowed values; and $f(5)$ has $4 - m$ values.

This gives $(4-m)^m(3-m)^{4-m}(4-m)$: $m = 0$: $4 \cdot 3^4 = 324$; $m = 1$: $3^2 \cdot 2^3 = 72$; $m = 2$: $2^3 \cdot 1^2 = 8$; $m = 3$: $1 \cdot 0 = 0$; $m = 4$: 0.

Favourable $= 324 - \binom{4}{1}72 + \binom{4}{2}8 - 0 + 0 = 324 - 288 + 48 = 84$.

Probability $= \dfrac{84}{240} = \dfrac{7}{20} = 0.35$.

Final Answer: 0.35

Question 37

English

In a survey of 200 persons about three newspapers A, B and C, it was found that 25 persons read none of the three newspapers, 40 persons read exactly two of them and 12 persons read all three. Here $n(X)$ denotes the number of persons who read newspaper X, and $n(X \cap Y)$ the number who read both X and Y. Then the value of

$$n(A \cap B) + n(B \cap C) + n(C \cap A) - n(A) - n(B) - n(C)$$

is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

तीन समाचार-पत्रों A, B तथा C के विषय में 200 व्यक्तियों के एक सर्वेक्षण में पाया गया कि 25 व्यक्ति तीनों में से कोई समाचार-पत्र नहीं पढ़ते, 40 व्यक्ति इनमें से ठीक दो पढ़ते हैं तथा 12 व्यक्ति तीनों पढ़ते हैं। यहाँ $n(X)$ समाचार-पत्र X पढ़ने वाले व्यक्तियों की संख्या है, तथा $n(X \cap Y)$, X तथा Y दोनों पढ़ने वाले व्यक्तियों की संख्या है। तब

$$n(A \cap B) + n(B \cap C) + n(C \cap A) - n(A) - n(B) - n(C)$$

का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Split the 175 readers by how many papers they read: $e_3 = 12$ read all three, $e_2 = 40$ read exactly two, and $e_1 = 200 - 25 - 40 - 12 = 123$ read exactly one.

A person who reads exactly j of the papers is counted j times in $n(A)+n(B)+n(C)$ and $\binom{j}{2}$ times in $n(A \cap B) + n(B \cap C) + n(C \cap A)$ (once for each pair of papers that person reads).

So $n(A)+n(B)+n(C) = e_1+2e_2+3e_3 = 123+80+36 = 239$ and $n(A\cap B)+n(B\cap C)+n(C\cap A) = e_2 + 3e_3 = 40 + 36 = 76$.

Check with inclusion-exclusion: $n(A \cup B \cup C) = 239 - 76 + 12 = 175 = 200 - 25$.

Required value $= 76 - 239 = -163$. In general this expression equals $-(e_1 + e_2)$: a reader of exactly one or two papers contributes -1, and a reader of all three contributes $3 - 3 = 0$.

Final Answer: -163

Question 38

English

Let $N = 2^6 \cdot 3^6 \cdot 5^4 \cdot 7^2$. Then the number of ways of writing N as a product $a \cdot b$ of two even positive integers a and b, where $a \cdot b$ and $b \cdot a$ are regarded as the same, is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $N = 2^6 \cdot 3^6 \cdot 5^4 \cdot 7^2$ है। तब N को दो सम धनात्मक पूर्णांकों a तथा b के गुणनफल $a \cdot b$ के रूप में लिखने के तरीकों की संख्या, जहाँ $a \cdot b$ तथा $b \cdot a$ को समान माना जाता है, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

If a and b are even with $ab = N$, write $a = 2a'$, $b = 2b'$. Then $a'b' = \dfrac{N}{4} = M = 2^4 \cdot 3^6 \cdot 5^4 \cdot 7^2$, and conversely every factorisation $M = a'b'$ gives even $a = 2a'$, $b = 2b'$. So we count unordered factorisations of M.

Number of divisors: $d(M) = (4+1)(6+1)(4+1)(2+1) = 525$.

Ordered factorisations $M = a' \cdot b'$ correspond to divisors a': 525 of them. All exponents of M are even, so M is a perfect square with $\sqrt{M} = 2^2 \cdot 3^3 \cdot 5^2 \cdot 7 = 18900$, and exactly one factorisation has $a' = b'$.

The other 524 ordered factorisations pair off into 262 unordered ones, so the number of unordered factorisations is $262 + 1 = \dfrac{525+1}{2} = 263$.

Final Answer: 263

Question 39

English

Two numbers a and b are chosen independently and at random from the set of positive divisors of 48 (the same divisor may be chosen twice). Let p be the probability that the least common multiple of a and b is 48, and let q be the probability that a divides b. Then the value of $p - q$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

48 के धनात्मक भाजकों के समुच्चय में से दो संख्याएँ a तथा b स्वतन्त्र रूप से यादृच्छया चुनी जाती हैं (एक ही भाजक दो बार चुना जा सकता है)। माना कि p, a तथा b के लघुत्तम समापवर्त्य के 48 होने की प्रायिकता है, तथा q, a के b को विभाजित करने की प्रायिकता है। तब $p - q$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$48 = 2^4 \cdot 3$, so its divisors are $2^i 3^j$ with $0 \le i \le 4$, $0 \le j \le 1$: $d(48) = 5 \cdot 2 = 10$. Ordered pairs (a, b): 100, equally likely.

Write $a = 2^i 3^j$, $b = 2^k 3^l$. Then $\text{lcm}(a, b) = 2^{\max(i,k)} 3^{\max(j,l)}$.

$\text{lcm}(a, b) = 48$ iff $\max(i, k) = 4$ and $\max(j, l) = 1$. Pairs (i, k) in $\{0, \ldots, 4\}^2$ with maximum 4: $5^2 - 4^2 = 9$ (that is, $2 \cdot 4 + 1$). Pairs (j, l) with maximum 1: $2^2 - 1 = 3$. So 27 pairs and $p = \dfrac{27}{100}$.

$a \mid b$ iff $i \le k$ and $j \le l$. Pairs with $i \le k$ in $\{0, \ldots, 4\}$: $\binom{5}{2} + 5 = 15$; pairs with $j \le l$ in $\{0, 1\}$: 3. So 45 pairs and $q = \dfrac{45}{100}$.

$$p - q = \frac{27 - 45}{100} = -\frac{18}{100} = -0.18.$$

Final Answer: -0.18

Question 40

English

Let α be the number of ordered pairs (a, b) of positive integers such that the least common multiple of a and b is 60, and let β be the number of ways of writing 3600 as a product $a \cdot b$ of two positive integers, where $a \cdot b$ and $b \cdot a$ are regarded as the same. Then the value of $\beta - \alpha$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि α धनात्मक पूर्णांकों के उन क्रमित युग्मों (a, b) की संख्या है जिनका लघुत्तम समापवर्त्य 60 है, तथा β, 3600 को दो धनात्मक पूर्णांकों के गुणनफल $a \cdot b$ के रूप में लिखने के तरीकों की संख्या है, जहाँ $a \cdot b$ तथा $b \cdot a$ को समान माना जाता है। तब $\beta - \alpha$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

α: write $a = 2^i 3^j 5^k$, $b = 2^{i'} 3^{j'} 5^{k'}$. $\text{lcm}(a, b) = 60 = 2^2 \cdot 3 \cdot 5$ iff $\max(i, i') = 2$, $\max(j, j') = 1$, $\max(k, k') = 1$ (and no other primes occur). For a prime with exponent e, the pairs with maximum e number $(e + 1)^2 - e^2 = 2e + 1$. So $\alpha = 5 \cdot 3 \cdot 3 = 45$.

β: $3600 = 2^4 \cdot 3^2 \cdot 5^2$ has $d(3600) = (4 + 1)(2 + 1)(2 + 1) = 45$ divisors, each giving an ordered factorisation $a \cdot \dfrac{3600}{a}$.

$3600 = 60^2$ is a perfect square, so one factorisation is $60 \cdot 60$ and the remaining 44 ordered ones pair off. Hence $\beta = \dfrac{45 + 1}{2} = 23$.

$\beta - \alpha = 23 - 45 = -22$.

Note: $\alpha = d(60^2)$ in general ($\prod(2e + 1)$ both ways), so $\beta = \dfrac{\alpha + 1}{2}$.

Final Answer: -22

Question 41

English

A convex polygon has 27 diagonals, and no three of its diagonals pass through a common point inside the polygon. Each diagonal is divided into segments by the points at which it meets the other diagonals inside the polygon. Then the total number of these segments, taken over all the diagonals, is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक उत्तल बहुभुज के 27 विकर्ण हैं, और इसके कोई तीन विकर्ण बहुभुज के अन्दर किसी उभयनिष्ठ बिन्दु से नहीं गुजरते। प्रत्येक विकर्ण उन बिन्दुओं द्वारा खण्डों में विभाजित होता है जिन पर वह बहुभुज के अन्दर अन्य विकर्णों से मिलता है। तब सभी विकर्णों पर लिए गए इन खण्डों की कुल संख्या _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For a convex n-gon the number of diagonals is $\binom{n}{2} - n = \dfrac{n(n-3)}{2}$. So $\dfrac{n(n-3)}{2} = 27$, i.e. $n^2 - 3n - 54 = 0 = (n-9)(n+6)$, and $n = 9$.

Two diagonals cross inside a convex polygon exactly when their four endpoints are distinct and alternate around the polygon; each set of 4 vertices determines exactly one such pair. Since no three diagonals are concurrent, distinct sets give distinct points: there are $\binom{9}{4} = 126$ interior intersection points.

If a diagonal contains k interior intersection points, they cut it into $k+1$ segments. Summing over all diagonals: total segments $= 27 + \sum k$.

Each interior intersection point lies on exactly 2 diagonals, so $\sum k = 2 \cdot 126 = 252$.

Total number of segments $= 27 + 252 = 279$.

Final Answer: 279

Question 42

English

Let $A = (0, 0)$, $B = (6, 0)$ and $C = (2, 0)$, and let $D \neq C$ be the point on the line AB such that $\frac{DA}{DB} = \frac{CA}{CB}$. Let $P = (0, 3)$ and $Q = (2, 4)$. Three of the six points A, B, C, D, P, Q are chosen at random. Then the probability that the three chosen points are the vertices of a triangle is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $A = (0, 0)$, $B = (6, 0)$ तथा $C = (2, 0)$ हैं, और माना कि $D \neq C$ रेखा AB पर वह बिन्दु है जिसके लिए $\frac{DA}{DB} = \frac{CA}{CB}$ है। माना कि $P = (0, 3)$ तथा $Q = (2, 4)$ हैं। छः बिन्दुओं A, B, C, D, P, Q में से तीन बिन्दु यादृच्छया चुने जाते हैं। तब चुने गए तीनों बिन्दुओं के एक त्रिभुज के शीर्ष होने की प्रायिकता _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

By the distance formula, $CA = 2$ and $CB = 4$, so C divides AB internally in the ratio $1 : 2$. A point $D = (x, 0)$ of line AB with $\frac{DA}{DB} = \frac{1}{2}$ satisfies $2|x| = |x - 6|$, i.e. $x = 2$ (the point C) or $x = -6$.

So $D = (-6, 0)$, the point dividing AB externally in the ratio $1 : 2$ (section formula: $\frac{1 \cdot (6, 0) - 2 \cdot (0, 0)}{1 - 2} = (-6, 0)$); D is the harmonic conjugate of C with respect to A and B.

Three points are vertices of a triangle unless they are collinear. Total choices: $\binom{6}{3} = 20$.

A, B, C, D all lie on the x-axis: $\binom{4}{3} = 4$ collinear triples.

P and Q are not on the x-axis, so a triple containing one of them and two axis points is never collinear. Triples with both P and Q: the line PQ has slope $\frac{4 - 3}{2 - 0} = \frac{1}{2}$ and equation $y = \frac{x}{2} + 3$, which meets the x-axis at $(-6, 0) = D$ only. So D, P, Q is the only such collinear triple.

Collinear triples: $4 + 1 = 5$; triangles: 15. Probability $= \frac{15}{20} = 0.75$.

Final Answer: 0.75

Question 43

English

Ten lines are drawn in a plane so that no two of them are parallel and no three of them pass through a common point. Let B and U be the numbers of bounded and unbounded regions, respectively, into which these lines divide the plane. Then the value of $\frac{B}{U}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक तल में दस रेखाएँ इस प्रकार खींची जाती हैं कि उनमें से कोई दो समान्तर नहीं हैं और कोई तीन एक उभयनिष्ठ बिन्दु से नहीं गुजरतीं। माना कि ये रेखाएँ तल को जितने परिबद्ध तथा अपरिबद्ध क्षेत्रों में विभाजित करती हैं, उनकी संख्याएँ क्रमशः B तथा U हैं। तब $\frac{B}{U}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Add the lines one at a time. The k-th line meets the previous $k-1$ lines in $k-1$ distinct points (no two parallel, no three concurrent), so it is divided into k pieces, and each piece splits one existing region into two: the k-th line adds k regions.

Starting from 1 region: $R = 1 + (1 + 2 + \cdots + 10) = 1 + 55 = 56$. Equivalently $R = 1 + n + \binom{n}{2}$ with $\binom{10}{2} = 45$ points of intersection.

Draw a circle large enough to contain all 45 points of intersection. Each line crosses it twice, giving 20 points on the circle and 20 arcs. Outside the circle no two lines meet, so each unbounded region contains exactly one arc, and each arc lies in one unbounded region: $U = 20$.

Hence $B = R - U = 56 - 20 = 36$ (equal to $\binom{10-1}{2}$).

$\frac{B}{U} = \frac{36}{20} = 1.8$.

Final Answer: 1.80

Question 44

English

A 4×4 board is divided into 16 unit squares by grid lines. A rectangle on the board is a rectangle whose sides lie along the grid lines (every square is also a rectangle). Then the number of rectangles on the board that contain none of the four central unit squares is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक 4×4 बोर्ड को ग्रिड रेखाओं द्वारा 16 इकाई वर्गों में विभाजित किया गया है। बोर्ड पर एक आयत वह आयत है जिसकी भुजाएँ ग्रिड रेखाओं पर स्थित हैं (प्रत्येक वर्ग भी एक आयत है)। तब बोर्ड पर उन आयतों की संख्या, जो चार केन्द्रीय इकाई वर्गों में से किसी को भी नहीं समाहित करते, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Bijection: a rectangle is fixed by choosing 2 of the 5 horizontal grid lines and 2 of the 5 vertical grid lines, i.e. by a row interval and a column interval. There are $\binom{5}{2} = 10$ of each, so 100 rectangles in all.

Number the rows and columns $1, 2, 3, 4$; the central squares are those in rows 2, 3 and columns 2, 3. A rectangle contains a central square exactly when its row interval contains row 2 or row 3 and its column interval contains column 2 or column 3 (then it contains the square at that row and column).

A row interval containing neither row 2 nor row 3 must lie in row 1 alone or row 4 alone: 2 intervals. So $10 - 2 = 8$ row intervals meet $\{2, 3\}$; likewise 8 column intervals.

Complementary counting: rectangles containing at least one central square: $8 \cdot 8 = 64$.

Rectangles containing none: $100 - 64 = 36$.

Final Answer: 36

Question 45

English

Three of ten chairs placed in a row are chosen at random. Let p be the probability that no two of the chosen chairs are adjacent, and let q be the probability that the three chosen chairs are consecutive. Then the value of $q - p$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक पंक्ति में रखी दस कुर्सियों में से तीन कुर्सियाँ यादृच्छया चुनी जाती हैं। माना कि p चुनी गई कुर्सियों में से किन्हीं दो के आसन्न न होने की प्रायिकता है, तथा q तीनों चुनी गई कुर्सियों के क्रमागत होने की प्रायिकता है। तब $q - p$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

All $\binom{10}{3} = 120$ choices are equally likely.

No two chosen chairs adjacent: place the 7 unchosen chairs in a row; they form 8 gaps (including both ends), and choosing 3 different gaps for the chosen chairs gives exactly the choices with no two adjacent. Number: $\binom{8}{3} = 56$, so $p = \dfrac{56}{120}$.

Three consecutive chairs: the block can start at chair $1, 2, \ldots, 8$, so 8 choices and $q = \dfrac{8}{120}$.

$q - p = \dfrac{8 - 56}{120} = -\dfrac{48}{120} = -0.4.$

Final Answer: -0.40

Question 46

English

A college has 5 teachers in Physics, 4 in Chemistry and 3 in Mathematics. A committee of 4 teachers is to be formed containing at least one teacher from each of the three departments. Two of the teachers, X (from Physics) and Y (from Mathematics), refuse to serve together on the committee. Then the number of ways of forming the committee is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक महाविद्यालय में भौतिकी के 5, रसायन के 4 तथा गणित के 3 शिक्षक हैं। 4 शिक्षकों की एक समिति बनानी है जिसमें तीनों विभागों में से प्रत्येक से कम से कम एक शिक्षक हो। शिक्षकों में से दो, X (भौतिकी से) तथा Y (गणित से), समिति में एक साथ कार्य करने से मना करते हैं। तब समिति बनाने के तरीकों की संख्या _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Committees of 4 with all three departments: the department sizes are $2, 1, 1$ in some order. Physics twice: $\binom{5}{2} \cdot 4 \cdot 3 = 120$; Chemistry twice: $5 \cdot \binom{4}{2} \cdot 3 = 90$; Mathematics twice: $5 \cdot 4 \cdot \binom{3}{2} = 60$. Total 270.

(Check by complementary counting: $\binom{12}{4} - \left[\binom{7}{4} + \binom{8}{4} + \binom{9}{4}\right] + \left[\binom{3}{4} + \binom{4}{4} + \binom{5}{4}\right] = 495 - 231 + 6 = 270$.)

Remove the committees containing both X and Y. With X and Y in, Physics and Mathematics are covered, and the other 2 members come from the remaining 4 Physics, 4 Chemistry and 2 Mathematics teachers, with at least one chemist.

By complementary counting, that is $\binom{10}{2} - \binom{6}{2} = 45 - 15 = 30$.
Number of committees $= 270 - 30 = 240$.

Final Answer: 240

Question 47

English

Let λ be the real number such that

$$\sum_{r=0}^{13} \left(r^2 + \lambda r\right) \binom{13}{r}^2 = 0.$$

Then the value of λ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि λ वह वास्तविक संख्या है जिसके लिए

$$\sum_{r=0}^{13} \left(r^2 + \lambda r\right) \binom{13}{r}^2 = 0$$

है। तब λ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $n = 13$. Use $r\binom{n}{r} = n\binom{n-1}{r-1}$ (the result behind the differentiation technique $\dfrac{d}{dx}(1+x)^n = n(1+x)^{n-1}$) and the symmetry $\binom{n}{r} = \binom{n}{n-r}$.
$\sum_r r\binom{n}{r}^2 = \sum_r n\binom{n-1}{r-1}\binom{n}{n-r} = n\binom{2n-1}{n-1}$ by Vandermonde (lower indices add to $n-1$; equivalently the coefficient of x^{n-1} in $(1+x)^{n-1}(1+x)^n$).
$\sum_r r^2\binom{n}{r}^2 = \sum_r \left(r\binom{n}{r}\right)^2 = n^2 \sum_r \binom{n-1}{r-1}^2 = n^2\binom{2n-2}{n-1}$.
The condition is $n^2\binom{2n-2}{n-1} + \lambda\, n\binom{2n-1}{n-1} = 0$. Since $\binom{2n-1}{n-1} = \dfrac{2n-1}{n}\binom{2n-2}{n-1}$, this gives $n^2 + \lambda(2n-1) = 0$, i.e. $\lambda = -\dfrac{n^2}{2n-1}$.
For $n = 13$: $\lambda = -\dfrac{169}{25} = -6.76$.

Final Answer: -6.76

Question 48

English

Let D, E and F be the midpoints of the sides BC, CA and AB, respectively, of a triangle ABC. Three of the six points A, B, C, D, E, F are chosen at random. Then the probability that the three chosen points are the vertices of a triangle whose area is one-fourth of the area of triangle ABC is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि D, E तथा F क्रमशः त्रिभुज ABC की भुजाओं BC, CA तथा AB के मध्य-बिन्दु हैं। छः बिन्दुओं A, B, C, D, E, F में से तीन बिन्दु यादृच्छया चुने जाते हैं। तब चुने गए तीनों बिन्दुओं के एक ऐसे त्रिभुज के शीर्ष होने की प्रायिकता, जिसका क्षेत्रफल त्रिभुज ABC के क्षेत्रफल का एक-चौथाई है, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

There are $\binom{6}{3} = 20$ choices. Let Δ be the area of triangle ABC. Classify by the number of vertices of ABC chosen.

Three vertices: triangle ABC, area Δ (1 choice).

Two vertices and one midpoint ($3 \cdot 3 = 9$ choices): if the midpoint is on the side joining the two vertices (e.g. A, B, F), the points are collinear (3 choices); otherwise (e.g. A, B, D) the triangle has half the base or half the height of a triangle of area Δ, e.g. $[ABD] = \frac{1}{2}[ABC]$, area $\frac{\Delta}{2}$ (6 choices).

One vertex and two midpoints ($3 \cdot 3 = 9$ choices), say vertex A: $[AEF] = \frac{\Delta}{4}$ (corner triangle); $[ADE] = \frac{1}{2}[ADC] = \frac{1}{2} \cdot \frac{\Delta}{2} = \frac{\Delta}{4}$ since E is the midpoint of AC; similarly $[ADF] = \frac{\Delta}{4}$. By symmetry all 9 have area $\frac{\Delta}{4}$.

Three midpoints: the medial triangle DEF, area $\frac{\Delta}{4}$ (1 choice).

Favourable: $9 + 1 = 10$; probability $= \frac{10}{20} = 0.5$.

Final Answer: 0.50

Question 49

English

From a 4×4 board of unit squares, the unit square at one corner is removed. A rectangle in the remaining figure is a rectangle with sides along the grid lines that is made up only of unit squares of the remaining figure. Then the number of rectangles in the remaining figure that are not squares is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

इकाई वर्गों के एक 4×4 बोर्ड से एक कोने का इकाई वर्ग हटा दिया जाता है। शेष आकृति में एक आयत वह आयत है जिसकी भुजाएँ ग्रिड रेखाओं पर हैं तथा जो केवल शेष आकृति के इकाई वर्गों से बना है। तब शेष आकृति में उन आयतों की संख्या, जो वर्ग नहीं हैं, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

On the full 4×4 board a rectangle is fixed by 2 of the 5 horizontal and 2 of the 5 vertical grid lines: $\binom{5}{2}^2 = 100$ rectangles. Squares of side k number $(5-k)^2$, so there are $16 + 9 + 4 + 1 = 30$ squares and 70 non-square rectangles.

A rectangle lies in the remaining figure iff it does not contain the removed corner square. A rectangle contains that square iff its row interval starts at the first row and its column interval starts at the first column: $4 \cdot 4 = 16$ rectangles.

Among these 16, the squares are those with equal row and column lengths: sizes $1, 2, 3, 4$, i.e. 4 squares; so 12 are non-squares.

Non-square rectangles in the remaining figure: $70 - 12 = 58$. (Similarly there are $100 - 16 = 84$ rectangles and $30 - 4 = 26$ squares in it.)

Final Answer: 58

Question 50

English

The value of $\sum_{d \mid 180} (-1)^{d+1}\, d$, where the sum is taken over all positive divisors d of 180, is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\sum_{d \mid 180} (-1)^{d+1}\, d$ का मान, जहाँ योग 180 के सभी धनात्मक भाजकों d पर लिया गया है, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For odd d, $(-1)^{d+1} = 1$, and for even d, $(-1)^{d+1} = -1$. So the sum equals (sum of odd divisors) − (sum of even divisors).

$180 = 2^2 \cdot 3^2 \cdot 5$. Every divisor is $2^a m$ with $a \in \{0, 1, 2\}$ and m an odd divisor, i.e. a divisor of $3^2 \cdot 5 = 45$.

Sum of odd divisors ($a = 0$): $(1 + 3 + 9)(1 + 5) = 13 \cdot 6 = 78$.

Sum of even divisors ($a = 1, 2$): $(2+4) \cdot 78 = 468$. (Check: $\sigma(180) = (1+2+4) \cdot 78 = 546 = 78+468$.)

The sum is $78 - 468 = -390$, i.e. $(1 - 2 - 4) \cdot 78$.

Final Answer: -390